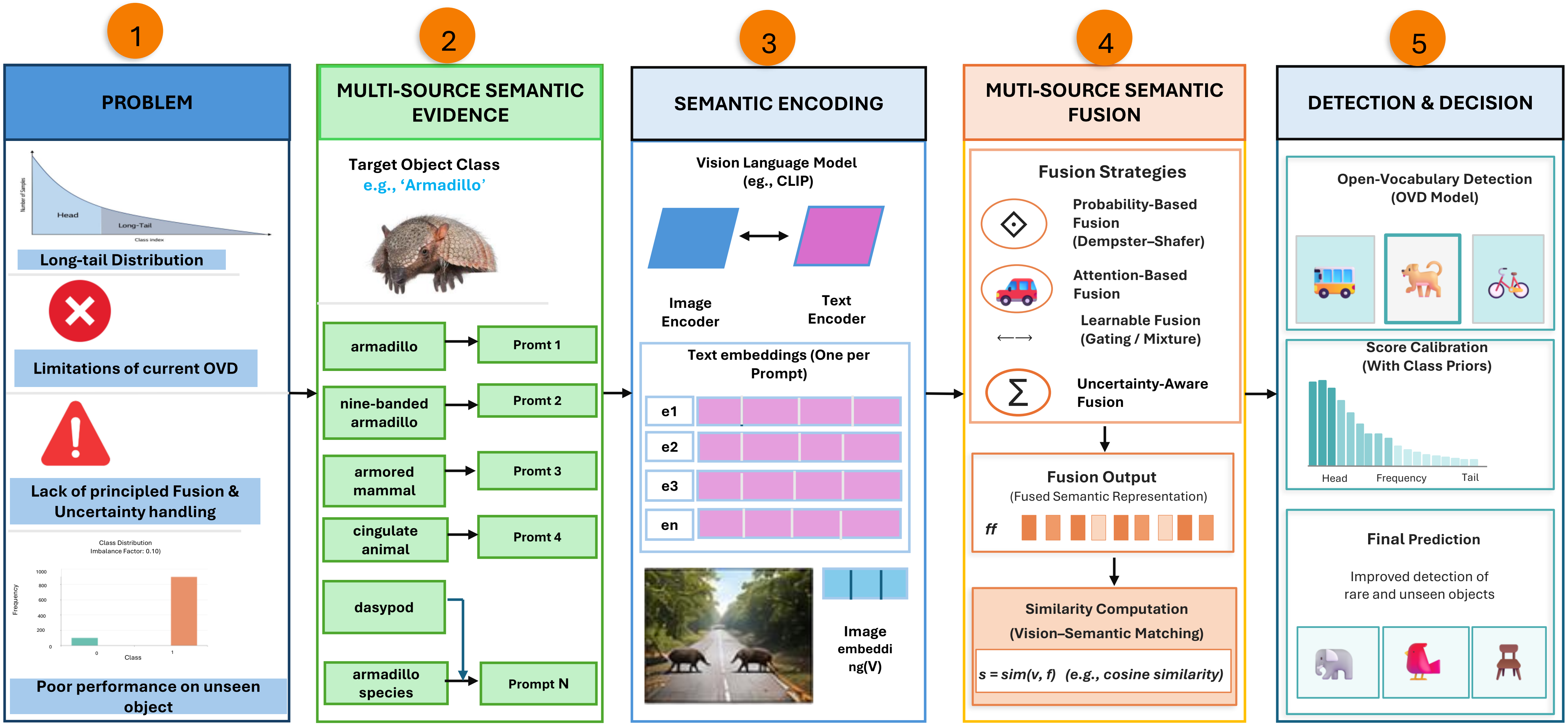
Multi-Source Semantic Fusion for Long-Tail Object Detection — Introduction



EXPECTED IMPACT

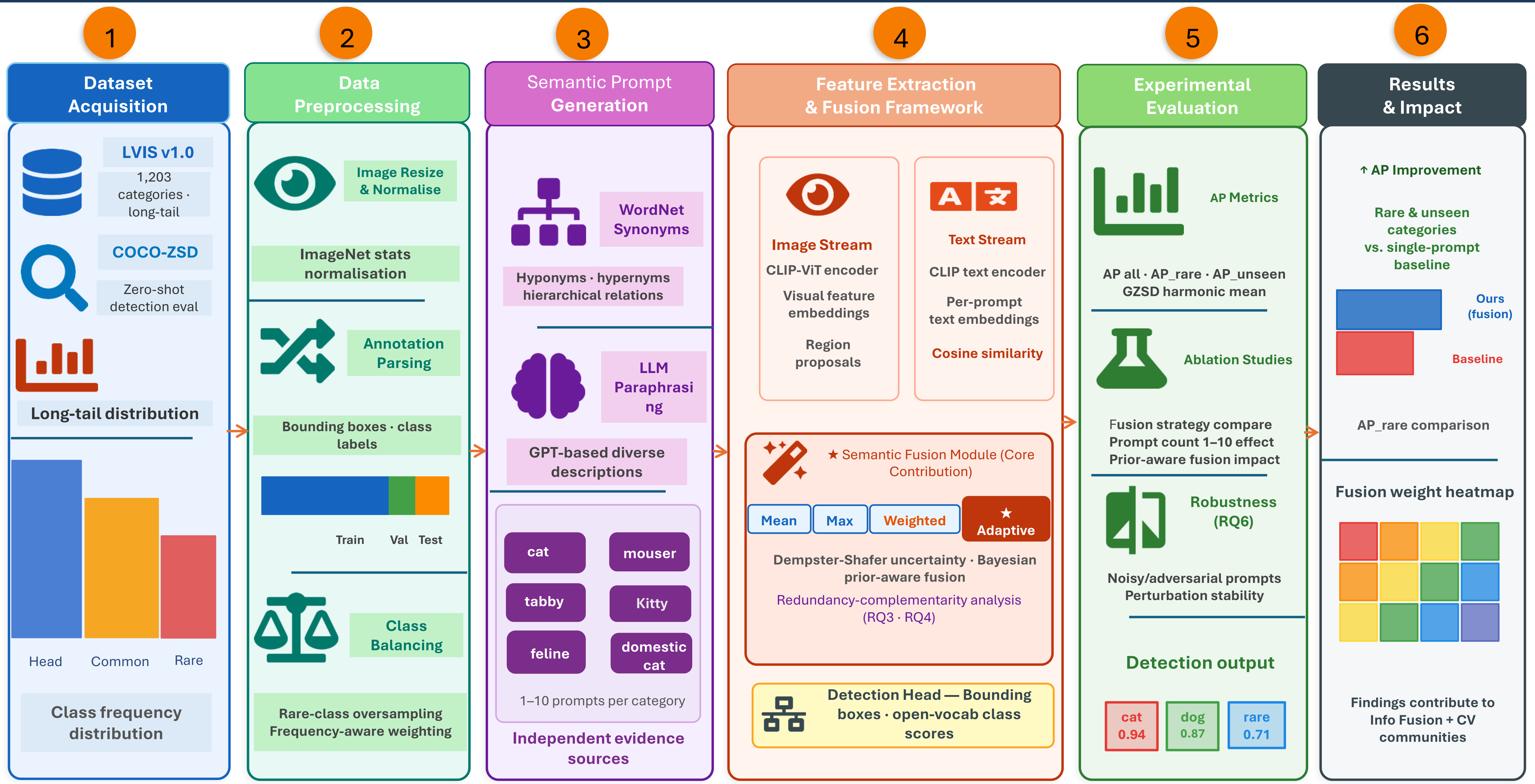
Better detection of rare & unseen objects

Robust to prompt variations & uncertainty

Principled fusion of semantic evidence

Advances in Information Fusion & OVD

Multi-Source Semantic Fusion for Long-Tail Object Detection — Methodology Workflow



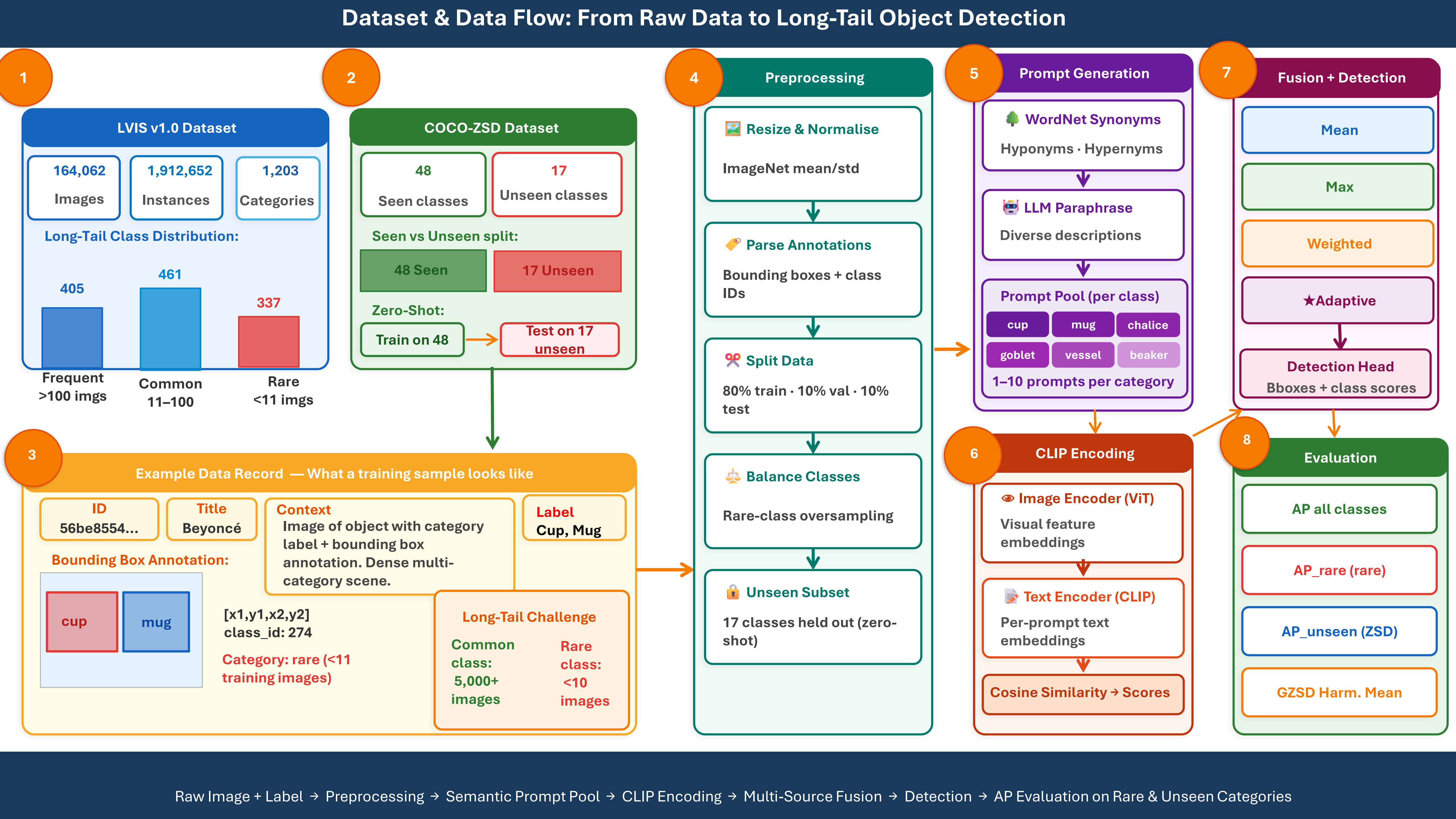
LVIS v1.0 & COCO-ZSD — Dataset Feature Structure & Sample Records

Image & Visual Features
Annotation & Spatial Features
Zero-Shot & Semantic Features
Dataset Metadata (freq_group)
Target Label

image_id	img_w	img_h	category_id	bbox_x	bbox_y	bbox_w	bbox_h	area	freq_group	prompt_count	cos_sim	seen_cls	unseen	AP_rare	label
56be8554	1280	720	274	312	128	64	89	5696	rare	7	0.823	1	0	0.12	0
a14008c9	1024	768	47	64	200	128	160	20480	freq	3	0.941	1	0	0.89	1
56cc59cd	1920	1080	891	450	300	220	180	39600	common	5	0.762	0	1	0.34	0
56ce792c	640	480	103	80	120	100	140	14000	rare	9	0.688	1	0	0.09	0
5a7deb71	1280	720	1102	200	150	300	250	75000	unseen	10	0.594	0	1	0.71	1

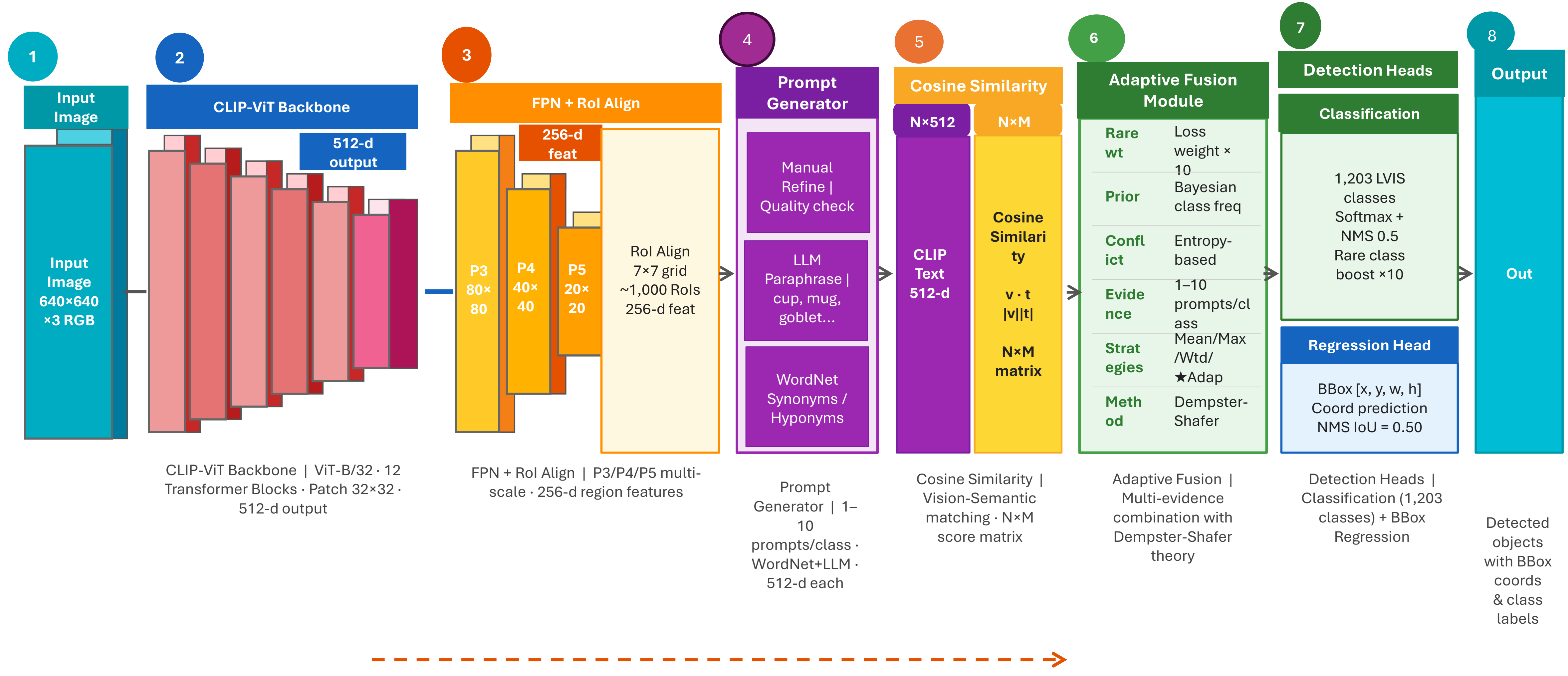
image_id Unique Image ID
img_w · img_h Image Dimensions (px)
category_id LVIS Category Index
bbox_x BBox X-coordinate
bbox_y BBox Y-coordinate
bbox_w BBox Width
bbox_h BBox Height
area Segment Area
freq_group Head/Common /Rare/Unseen
prompt count Prompts per Class (1–10)
cos_sim CLIP Cosine Score
seen_cls Seen Class Flag (0/1)
unseen Unseen Flag (0/1)
AP_rare Detection AP Score
label Target (1=detect, 0=miss)

LVIS v1.0: 164,062 images · 1,203 categories · 1,912,652 instances · 337 rare classes (<11 images) | COCO-ZSD: 48 seen + 17 unseen classes · Zero-Shot Detection benchmark



Dataset & Data Flow: From Raw Data to Long-Tail Object Detection
1
LVIS v1.0 Dataset
164,062
Images
1,912,652
Instances
1,203
Categories
Long-Tail Class Distribution:
405
461
337
Frequent
>100 imgs
Common
11–100
Rare
<11 imgs
2
COCO-ZSD Dataset
48
Seen classes
17
Unseen classes
Seen vs Unseen split:
48 Seen
17 Unseen
Zero-Shot:
Train on 48
Test on 17 unseen
3
Example Data Record — What a training sample looks like
ID
56be8554...
Title
Beyoncé
Context
Image of object with category label + bounding box annotation. Dense multi-category scene.
Label
Cup, Mug
Bounding Box Annotation:
cup
mug
[x1,y1,x2,y2]
class_id: 274
Category: rare (<11 training images)
Long-Tail Challenge
Common class: 5,000+ images
Rare class: <10 images
4
Preprocessing
Resize & Normalise
ImageNet mean/std
Parse Annotations
Bounding boxes + class IDs
Split Data
80% train · 10% val · 10% test
Balance Classes
Rare-class oversampling
Unseen Subset
17 classes held out (zero-shot)
5
Prompt Generation
WordNet Synonyms
Hyponyms · Hypernyms
LLM Paraphrase
Diverse descriptions
Prompt Pool (per class)
cup
mug
chalice
goblet
vessel
beaker
1–10 prompts per category
6
CLIP Encoding
Image Encoder (ViT)
Visual feature embeddings
Text Encoder (CLIP)
Per-prompt text embeddings
Cosine Similarity → Scores
7
Fusion + Detection
Mean
Max
Weighted
★Adaptive
Detection Head
Bboxes + class scores
8
Evaluation
AP all classes
AP_rare (rare)
AP_unseen (ZSD)
GZSD Harm. Mean
Raw Image + Label → Preprocessing → Semantic Prompt Pool → CLIP Encoding → Multi-Source Fusion → Detection → AP Evaluation on Rare & Unseen Categories

Network Architecture 1 — LVIS v1.0: Long-Tail Object Detection with Multi-Source Adaptive Semantic Fusion



Skip connection: FPN P3 region features → Adaptive Fusion Module

Network Architecture 2 — COCO-ZSD: Zero-Shot Detection with CLIP Semantic Transfer & GZSD Evaluation